HW4 Due 11/1

geo ① 1) $p = 0.1$ $q = 1 - 0.1 = 0.9$ a sucess is missing, so switch p + q

$P(X \geq 13) = 1 - P(X < 13)$

$= 1 - \sum_{x=1}^{12} (1-p)^{x-1} p$ by def. of cdf

$= 1 - \sum_{x=1}^{12} (0.9)^{x-1}(0.1)$

$= 1 - 0.1 \sum_{x=1}^{12} (0.9)^{x-1}$ $\quad \frac{1-r^n}{1-r}$

$= 1 - 0.1 \sum_{x=0}^{11} (0.9)^{x} = 1 - 0.1\left(\frac{1-0.9^{12}}{1-0.9}\right) = 0.282$

nbinom

2) makes 2 misses in 1st 29 shots, and 30th

$P(X = 30) = \binom{29}{2} \cdot 0.1^3 \cdot 0.9^{27} = 406 \cdot 0.001 \cdot 0.0581 = 0.0235886$

② 1) $\mu = \frac{r}{p} = \frac{10}{0.6} = 16.667$

$\sigma^2 = \frac{10(1-0.6)}{(0.6)^2} = \frac{4}{0.36} = 11.11$

$SD = \sqrt{\sigma^2} = \sqrt{11.11} = 3.33$

2) $P(X=16) = \binom{15}{9} \cdot 0.6^{10} \cdot 0.4^{6} = 5005 \cdot 0.006 \cdot 0.004 = 0.12012$

③ 11/hr $\quad P(X > 10) = 1 - P(X \leq 10)$

$\lambda = 11$ $\quad = 1 - \sum_{X=0}^{10} \frac{11^x e^{-11}}{x!}$

$= 1 - e^{-11} \sum_{X=0}^{10} \frac{11^x}{x!}$

④ $E[(X^2 - 2\mu X + \mu^2)(X-\mu)] = E[X^3 - 2\mu X^2 + \mu^2 X - \mu X^2 + 2\mu^2 X - \mu^3] =$

bc Poisson

$E[X^3 - 3\mu X^2 + 3\mu^2 X - \mu^3] = E[X^3] - 3\mu E[X^2] + 3\mu^2 E[X] - \mu^3$

$\sigma^2 = \mu = \lambda = E[X]$ $\quad E[X^3] - 3\lambda E[X^2] + 3\lambda^3 - \lambda^3 = E[X^3] - 3\lambda E[X^2] + 2\lambda^3$

$\sigma^2 = E[X^2]$ ① $\lambda = E[X^2] - (E[X])^2$ ② $M(t) = e^{\lambda(e^t-1)}$

$\lambda = E[X^2] - \lambda^2$ $\quad M'(t) = \lambda e^t e^{\lambda(e^t-1)}$

$E[X^2] = \lambda^2 + \lambda$ $\quad M''(t) = \lambda e^t e^{\lambda(e^t-1)}(\lambda e^t) + \lambda e^t e^{\lambda(e^t-1)}$

$M''(t) = \lambda e^{\lambda(e^t-1)}(\lambda e^{2t} + e^t)$

$M'''(t) = \lambda e^{\lambda(e^t-1)}(2\lambda e^{2t} + e^t) + (\lambda e^{2t} + e^t)(\lambda e^{\lambda(e^t-1)} e^t)$

$E[X^3] = M'''(0) = \lambda(2\lambda+1) + (\lambda+1)(\lambda) = 2\lambda^2 + \lambda + \lambda^2 + \lambda = 3\lambda^2 + 2\lambda$

$M''(t) = \lambda(\lambda e^{2t} + e^t) e^{\lambda(e^t-1)}$

$M'''(t) = \lambda\left[\lambda(e^{2t}\cdot 2 + e^t) e^{\lambda(e^t-1)} + (\lambda e^{2t} + e^t)(e^{\lambda(e^t-1)})(\lambda e^t)\right]$

$\lambda\left[(2\lambda e^{2t} + e^t)(e^{\lambda(e^t-1)}) + (\lambda e^{2t} + e^t)(e^{\lambda(e^t-1)})(\lambda e^t)\right]$

$M'''(0) = \lambda\left[(2\lambda+1)(1) + (\lambda+1)(1)(\lambda)\right] = \lambda\left[(2\lambda+1) + (\lambda^2+\lambda)\right] =$

$2\lambda^2 + \lambda + \lambda^3 + \lambda^2 =$

$\lambda^3 + 3\lambda^2 + \lambda$

⑤ $f(x) = c(x - x^2) \quad 0 < x < 1$

1) $P(0 < x < 1) = \int_0^1 c(x - x^2)\,dx$

$\int_0^1 x\,dx - \int_0^1 x^2\,dx = \left[\frac{x^2}{2}\right]_0^1 - \left[\frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$

$c \cdot \frac{1}{6} = 1$

$c = 6$

2) $P(0.3 < X < 0.6) = \int_{0.3}^{0.6} 6(x - x^2)\,dx = 6\int_{0.3}^{0.6} (x - x^2)\,dx =$

$6\left[\frac{x^2}{2} - \frac{x^3}{3}\right]_{0.3}^{0.6} = 6\left[\left(\frac{0.6^2}{2} - \frac{0.6^3}{3}\right) - \left(\frac{0.3^2}{2} - \frac{0.3^3}{3}\right)\right] =$

$6((0.18 - 0.072) - (0.045 - 0.009)) =$

$6(0.108 - 0.036) = 0.432$

⑥ $F(x) = (1 + e^{-x})^{-1}$

$F'(x) = -1(1 + e^{-x})^{-2}(-e^{-x}) = \frac{1}{e^x(1 + e^{-x})^2}$

$f(x) = \frac{1}{e^x(1 + e^{-x})^2} \quad \text{for } -\infty < x < \infty$

4 cont. $E[X^3] - 3\lambda E[X^2] + 2\lambda^3 =$

$= \lambda^3 + 3\lambda^2 + \lambda - 3\lambda(\lambda^2 + \lambda) + 2\lambda^3 = \lambda^3 + 3\lambda^2 + \lambda - 3\lambda^3 - 3\lambda^2 + 2\lambda^3$

$= \boxed{\lambda}$

$E[(X-\mu)^3] = \lambda$

$\sigma^3 = \sigma^2 \cdot \sigma = \lambda \cdot \sqrt{\lambda} = (\lambda)^{3/2}$ $\quad \frac{E[(X-\mu)^3]}{\sigma^3} = \frac{\lambda}{\lambda^{3/2}} = \boxed{\frac{1}{\sqrt{\lambda}}}$

As λ increases, the index of skewness becomes smaller and approaches 0.